



Smart Device Services Manager

HP Print Quality Diagnostics - Technical Details

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Document Overview

HP Print Quality Diagnostics is a suite of solutions integrated into HP SDS Manager and device firmware allowing for remote and/or user-interactive print quality monitoring.

HP Print Quality Diagnostics will be released via hardware and software. The hardware, called an inline scan bar, which either comes pre-integrated into the device, or is available when adding an optional paper handling accessory. This hardware can be used to remotely trigger a diagnostic check, which involves custom designed diagnostic pages automatically uploading to the SDS cloud for analysis. Alternatively, the device firmware will analyse its own ordinary printed pages, which then sends analysis data to the SDS cloud but no images or print content. This hardware will be available for a select number of recent HP models.

The software app is installed onto the device console. The app will also involve printing SDS generated diagnostic pages, and securely uploading the scans to the HP SDS cloud for analysis. The app will be available for a wider variety of models

Regardless of the analysis engine used, any issues found during the analysis will result in the generation of a new type of engineering advisory, "[PrintQualityDiagnostics](#)", which can be accessed via the existing HP SDS Manager engineering advisory system.



HP Print Quality Diagnostics Analysis Engines

Inline Scan Bar (Hardware) – Remote User Invoked Diagnostics

User invoked use of the Inline Scan Bar is performed via the SDS Manager Portal. This utilises HP's print-path Inline Scan Bar to capture scans of SDS-generated diagnostic test pages and forward them to the cloud for analysis. The diagnostic pages are designed to allow for the analysis of a broad range of print quality issues.

Due to this engine's use of SDS-generated diagnostic test pages, diagnostic runs that employ this engine are included in the Operation Audit Log. The scanned images of the diagnostic pages can also be obtained for each diagnostic run.

NOTE: User invoked use of the Inline Scan Bar requires that on premises JAMC's be at version 4.1.6492 or above. If a device is being managed by a JAMC below that level the operation will fail.

Inline Scan Bar will initially be available in the following product series:

- HP Color LaserJet Enterprise Flow MFP X677z+ (Default configuration)
- HP Color LaserJet Managed MFP E786* (Requires ILSB accessory)
- HP Color LaserJet Managed MFP E877* (Requires ILSB accessory)



Print Path Scanning via
Inline Scan Bar (ILSB) Accessory)



Print Path Scanning via
Built-in Inline Scan Bar (ILSB)

Inline Scan Bar (Hardware) – Firmware Invoked Diagnostics

Firmware invoked analysis also takes advantage of HP's Inline Scan Bar but is only available for the built in scan bar, not for the add-on accessory. This analysis engine differs from the remote user



invocation in that it analyses the non-printable leading-edge of ordinary printed pages to detect toner artifacts rather than custom-designed diagnostic pages. No page content is used in the engine analysis. In this use case, SDS receives only analytic data from the device. No images or print content leaves the printer. Therefore, firmware invoked analysis is not included in the Operation Audit Log and no diagnostic scan images can be obtained. This Inline Scan Bar Engine Analysis is triggered automatically by the firmware on an HP-proprietary cadence and cannot be adjusted or disabled.

Firmware use of the Inline Scan Bar will, initially, be available in the following product series:

- HP Color LaserJet Enterprise Flow MFP X677z+ (Default configuration)



Print Path Scanning via
Built-in Inline Scan Bar (ILSB)

The HP Print Quality Diagnostics App (Software) – Local User Invoked Diagnostics

The HP Print Quality Diagnostics App is an analysis engine designed to work with diagnostic images created by the Automatic Document Feeder of devices. This engine includes an On-Device Services control panel application that can be manually invoked either by service technicians or end-users. This application utilizes the device's Automatic Document Feeder to scan SDS-generated diagnostic test pages and securely forward the scans to the cloud for analysis.



Automatic
Document
Feeder

The HP Print Quality Diagnostics App is designed for existing fleet devices and for Enterprise-class devices that do not have Inline Scan Bar options available. Due to leveraging these device's automatic document feeders, human interaction is required. This engine is best suited for technicians triaging print quality issues in real-time or for resellers with trusted customers that are willing to set up a manual process to run the application on

an agreed cadence.



Due to this engine's use of SDS-generated diagnostic test pages, diagnostic runs that employ this engine are included in the Operation Audit Log. The scanned images of the diagnostic pages can also be obtained for each diagnostic run.

NOTE: Users can install the HP Print Quality app remotely from the SDS Manager portal, but the print quality diagnostic run can **only** be invoked from the HP Print Quality app via the device control panel.

HP Print Quality Diagnostics Security

HP Print Quality Diagnostics has been designed to provide a secure print quality monitoring and diagnostics environment. In addition to the standard API and TLS security requirements used by SDS for all APIs, HP Print Quality Diagnostics provides security by taking the following additional steps:

Every diagnostic run that uses HP custom-designed diagnostic pages includes globally unique QR code identifiers.

Upon the receipt, by the HP SDS cloud, of scanned images, the images are scanned for malware prior to persisting the files in a restricted and quarantined storage environment.

The quarantined images are then automatically analysed to ensure the following pre-conditions are met:

- Do **all** the scanned images contain the globally unique QR codes?
- Do **all** the QR codes on **all the pages** match with SDS' expectation of the diagnostic run?
- Did SDS receive all the expected scans?

Only if each condition above is satisfied are the scans moved out of the quarantine storage environment and analysed for print quality issues.

The SDS Manager Portal / Insight Portal only has access to scanned images from any diagnostic run which have successfully passed the above security checks and have been released from HP's quarantined storage environment. It does not have access to any quarantined images.

If any one of the conditions above fail, the entire image archive is immediately deleted. Deleted image archive files cannot be recovered.

NOTE: The steps above ensure that even if non-diagnostic pages are inadvertently scanned and sent to SDS, the entire diagnostic run is aborted, and all images associated with the run are unrecoverably deleted.

NOTE: In the unlikely event an unexpected fault takes place, and the image archive is left in the quarantined storage environment, it will be permanently deleted after a short, pre-set period.



PQD Analysis Audit Log Entries vs Engineering Advisories

- A single diagnostic analysis run may detect multiple types of defects, or no defects at all. Each defect detected will result in the generation of a separate engineering advisory, thus audit logs may contain zero, one, or more links to the engineering advisories which detail any defects found.
- User invoked diagnostic runs will always result in the creation of a new audit log entry.
- Firmware invoked diagnostic analysis is automated and does not involve the printing/scanning/uploading of diagnostic test pages and so will never result in the creation of an audit log entry.
- It is possible (and even likely) that multiple diagnostics engines and/or multiple diagnostic runs will detect the same print quality defect. For as long as the engineering advisory detailing the defect found remains open, it will simply be updated with the link to the audit log entry of any new diagnostic runs which detect the same defect, thus a single engineering advisory may contain zero, one or more audit log links.

Miscellaneous Notes

- It will take up to 24hrs after initial device discovery for the SDS Manager Portal to be able to report on a device's level of print quality diagnostics support.
- It will take up to 24hrs after the installation of an Inline Scan Bar Accessory for the SDS Manager Portal to be able to report on a device's changed level of print quality diagnostics support.
- Audit Log entries are available from HP's SDS cloud for 18 months after creation and are then deleted.
- User invoked diagnostics runs will only succeed if the SDS-generated diagnostic test pages are printed on plain white paper.